

School Math Workbook Targeted Mastery Worksheet (Set 11)

Module: Number & Algebra · Linear Equations · Plane Geometry & Problem Solving

Name: _____

Class: _____

Date: _____

Total: 100
pts

Score: _____

Section 1: Fill in the Blanks

2 pts each, 20 pts total

1. In the number 483.59, the digit '8' is in the _____ place, representing 8 _____; the digit '5' is in the _____ place, representing 5 _____.

2. Insert '>', '<', or '=' in each $\bigcirc$:

(1) $7.2 \times 0.9 \bigcirc 7.2 \div 0.9$

(2) $4.5 \div 0.1 \bigcirc 4.5 \times 10$

(3) $0.85 \times 1.02 \bigcirc 0.85$

(4) $3.6 \div 1.2 \bigcirc 3.6$

3. Convert 0.075 into a percentage: _____, and into a fraction in its simplest form: _____; convert $\frac{1}{8}$ into a decimal: _____, and into a percentage: _____.

4. Given that $54 \div 24 = 2.25$, apply place value and quotient rules to evaluate:

$5.4 \div 24 =$ _____

$54 \div 0.24 =$ _____

$0.54 \div 2.4 =$ _____

5. For the pure recurring decimal $0.\dot{8}\dot{2}\dot{5}$, the recurring period is _____; its first 8 decimal places written in expanded form are _____; the 100th digit after the decimal point is _____.

6. Complete the rounding table below:

Number	Round to nearest integer	Round to 1 decimal place	Round to 2 decimal places
4.9964	_____	_____	_____
0.7258	_____	_____	_____

7. An expression for "7 more than 4 times x " is _____. Simplify the expressions: $5x + 4 + x =$ _____; $n + n - m + m =$ _____.

8. Fill in the missing numbers or units:

$2.4 \text{ m}^2 =$ _____ dm^2

$350 \text{ cm}^2 =$ _____ dm^2

$1.5 \text{ h} =$ _____ min

$72 \text{ km/h} =$ _____ m/s

9. A large cake weighing 1.2 kg is divided equally among 8 students. Each student receives _____ of the whole cake (express as a fraction), which weighs _____ kg.

10. A parallelogram has two adjacent sides of lengths 10 cm and 6 cm. If one of its heights is 8 cm, the corresponding base is _____ cm, and the area of the parallelogram is _____ cm^2 .

Section 2: Calculations & Column Operations

20 pts total

1. Mental Math (1 pt each, 4 pts total):

(1) $0.9 + 0.1 =$ _____ (2) $5.4 \div 0.6 =$ _____ (3) $0.25 \times 0.4 =$ _____ (4) $1 - 0.35 =$ _____

2. Column Operations (Part ① 2 pts; Part ② include check 3 pts; Part ③ round to 2 d.p. 3 pts; 8 pts total):

① 3.75×2.4

[Ans: _____]

② $276.5 \div 7.9$ (with check)

[Ans: _____]

③ $40 \div 42$ (to 2 d.p.)

[Ans: _____]

3. Order of Operations & Convenient Calculations (2 pts each, 8 pts total):

① $(1.25 + 2.5) \times 4$

[Ans: _____]

② $0.5 \times 1.6 \times 12.5$

[Ans: _____]

③ $0.4 \times (3.2 - 0.8) \div 1.2$

[Ans: _____]

④ $7.45 \times 21 + 5.75 \times 14$

[Ans: _____]

Section 3: Solve Equations & Algebraic Application

20 pts total

1. Solve the following equations (3 pts each, 12 pts total):

(1) $4x + 7.5 = 23.5$

[$x =$ _____]

(2) $4x - 1.2 = x + 4.5$

[$x =$ _____]

(3) $3(2x - 1.5) = 15$

[$x =$ _____]

(4) $5.6 \div x = 0.8$

[$x =$ _____]

2. Simplify and Evaluate (4 pts):

When $x = 2.5$, evaluate the algebraic expression: $3(2x - 1) - 4x + 5$.

[Simplified form: _____ , Value: _____]

3. Ratio and Algebraic Relations (4 pts):

Given that two positive quantities C and D satisfy the ratio $C : D = 3 : 5$.

(1) Express D in terms of C : $D =$ _____

(2) If $C = 18$, then $D =$ _____ ; if $D = 40$, then $C =$ _____ .

Section 4: Geometry & Spatial Measurement

20 pts total

1. Triangle Area and Height (5 pts):

A right-angled triangle has legs of lengths 0.44 m and 0.15 m, and a hypotenuse of length 0.5 m.

(1) Calculate the area of this right-angled triangle in m^2 .

(2) Calculate the height perpendicular to the hypotenuse in m (round to 2 decimal places).

[(1) Area: _____ m^2 ; (2) Height: _____ m]

2. Reverse Calculation of Trapezoid (5 pts):

The area of a trapezoid is 63 cm^2 , its height is 7 cm, and its top base is 6.8 cm. Find the length of its bottom base in cm.

[Ans: Bottom base is _____ cm]

3. Area of a Composite Figure (10 pts):

A mechanical component's cross-section is formed by combining: a top triangle A_1 with base 15 cm and height 7 cm ; a middle rectangle A_2 of length 15 cm and width 8 cm; and a right-side right triangle A_3 with base 9 cm and height 8 cm.

(1) Find each component area: $A_1 =$ _____ cm^2 ; $A_2 =$ _____ cm^2 ; $A_3 =$ _____ cm^2 .

(2) Calculate the total cross-sectional area of the component in cm^2 (show full addition steps).

[Ans: Total area is _____ cm^2]

Section 5: Real-World Word Problems

5 pts each, 20 pts total

1. A workshop has 8.6 kg of sugar. A worker packs a portion equally into 5 identical glass jars, leaving 0.35 kg of sugar unpacked. How many kilograms of sugar are packed into each glass jar on average?

[Ans: Each jar contains _____ kg]

2. A cafeteria purchaser bought 14 kg of chicken at 5.75 dollars/kg, and 21 kg of beef at 7.45 dollars/kg.

(1) What was the total cost for both chicken and beef?

(2) How much more was spent on beef than on chicken?

[(1) Total cost: \$ _____ ; (2) Beef cost \$ _____ more]

3. An electric bus travels from City A to City B, covering a total distance of 276.5 km at a constant speed in 3.5 h. The total energy consumed was equivalent to 7.9 L of fuel.

(1) What was the average speed of the bus in km/h?

(2) On average, how many kilometers does the bus travel per equivalent liter of fuel? (Round to 1 decimal place)

[(1) Average speed: _____ km/h; (2) Distance per liter: _____ km]

4. A science club created three display bars. Bar 1 consists of 4 wooden rods of length x cm placed end-to-end. Bar 2 consists of 1 wooden rod of length x cm connected to a 4.5 cm metal connector. Bar 1 is 1.5 cm longer than Bar 2.

(1) Set up an equation in terms of x and solve for x .

(2) Bar 3 consists of 3 wooden rods of length y cm, and its total length is exactly equal to the length of Bar 1. Find the value of y .

[(1) x = _____ cm; (2) y = _____ cm]



【Teacher Reference Key & Full Step-by-Step Solutions】 (For Scoring & Guidance)

Section 1: Fill in the Blanks

1. Tens; tens; tenths; tenths (or 0.1).
2. (1) $<$ (multiplying by < 1 decreases value, dividing by < 1 increases value); (2) $=$ ($4.5 \div 0.1 = 45 = 4.5 \times 10$); (3) $>$; (4) $<$.
3. 7.5%; $\frac{3}{40}$; 0.125; 12.5%.
4. 0.225; 225; 0.225.
5. 825; 0.82582582; 8 (since $100 \div 3 = 33 \text{ R } 1$, the first digit of the cycle 825 is 8).
6. $4.9964 \rightarrow 5$; 5.0; 5.00. $0.7258 \rightarrow 1$; 0.7; 0.73.
7. $4x + 7$; $6x + 4$; $2n$.
8. 240; 3.5; 90; 20 ($72 \times 1000 \div 3600 = 20 \text{ m/s}$).
9. $\frac{1}{8}$; 0.15 kg ($1.2 \div 8 = 0.15$).
10. 6; 48 (the height 8 must pair with the shorter side 6, area $= 6 \times 8 = 48 \text{ cm}^2$).

Section 2: Calculations & Column Operations

1. (1) 1; (2) 9; (3) 0.1; (4) 0.65.
2. ① $3.75 \times 2.4 = 9$; ② $276.5 \div 7.9 = 35$ (Check: $35 \times 7.9 = 276.5$); ③ $40 \div 42 \approx 0.95$.
3. ① $(1.25 + 2.5) \times 4 = 1.25 \times 4 + 2.5 \times 4 = 5 + 10 = 15$;
② $0.5 \times 1.6 \times 12.5 = (0.5 \times 2) \times (0.8 \times 12.5) = 1 \times 10 = 10$;
③ $0.4 \times 2.4 \div 1.2 = 0.96 \div 1.2 = 0.8$;
④ $7.45 \times 21 + 5.75 \times 14 = 156.45 + 80.5 = 236.95$.

Section 3: Equations & Algebra

1. (1) $4x = 16 \implies x = 4$; (2) $3x = 5.7 \implies x = 1.9$; (3) $2x - 1.5 = 5 \implies 2x = 6.5 \implies x = 3.25$; (4) $x = 5.6 \div 0.8 \implies x = 7$.
2. Simplification: $3(2x - 1) - 4x + 5 = 6x - 3 - 4x + 5 = 2x + 2$. At $x = 2.5$: $2(2.5) + 2 = 7$.
3. (1) $D = \frac{5}{3}C$; (2) When $C = 18$, $D = 30$; when $D = 40$, $C = 24$.

Section 4: Geometry & Spatial Measurement

1. (1) Area $= 0.44 \times 0.15 \div 2 = 0.033 \text{ m}^2$;
(2) Height $= 0.033 \times 2 \div 0.5 = 0.132 \approx 0.13 \text{ m}$.
2. Bottom base $= 63 \times 2 \div 7 - 6.8 = 18 - 6.8 = 11.2 \text{ cm}$.
3. (1) $A_1 = 15 \times 7 \div 2 = 52.5 \text{ cm}^2$; $A_2 = 15 \times 8 = 120 \text{ cm}^2$; $A_3 = 9 \times 8 \div 2 = 36 \text{ cm}^2$.
(2) Total area $= 52.5 + 120 + 36 = 208.5 \text{ cm}^2$.

Section 5: Word Problems

1. $(8.6 - 0.35) \div 5 = 8.25 \div 5 = 1.65 \text{ kg}$.
2. (1) Total cost $= 14 \times 5.75 + 21 \times 7.45 = 80.5 + 156.45 = 236.95$ dollars (\$236.95);
(2) Difference $= 156.45 - 80.5 = 75.95$ dollars (\$75.95).
3. (1) Average speed $= 276.5 \div 3.5 = 79 \text{ km/h}$;
(2) Distance per liter $= 276.5 \div 7.9 = 35 \text{ km/L}$.
4. (1) Equation: $4x - (x + 4.5) = 1.5 \implies 3x - 4.5 = 1.5 \implies 3x = 6 \implies x = 2 \text{ cm}$.
(2) Total length of Bar 1 $= 4 \times 2 = 8 \text{ cm}$; $3y = 8 \implies y = \frac{8}{3} \text{ cm}$ (or $2\frac{2}{3} \text{ cm}$).